

Section: 3.5 – STRENGTH OF COMPOSITE SANDWICH PANELS

Chapter: 3501

Title: SANDWICH PANELS (INTRO)

Revision: 0.1

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1 INTRODUCTION

1.1 OBJECT

This chapter is the first of a series dedicated to define the stress methodologies required for performing the sizing and strength check of composite sandwiches.

A sandwich structure is comprised of layered composite materials formed by bonding two or more thin facings or facesheets to a relatively thick core material. In this type of construction the facings resist nearly all of the inplane loads and out-of-plane bending moments. The thin facings provide nearly all of the bending stiffness because they are generally of a much higher modulus material and are located at the greatest distance from the neutral axis of the component. The core material provides spacing and transmits shear between the facesheets so that they are effective in bending about a common neutral axis. The core also provides most of the through-thickness shear rigidity of the sandwich construction and stabilizes the facesheets so they can be loaded to stress levels higher than the limits imposed by general buckling of thin plates (extract from S. Ward and L. Gintert article).

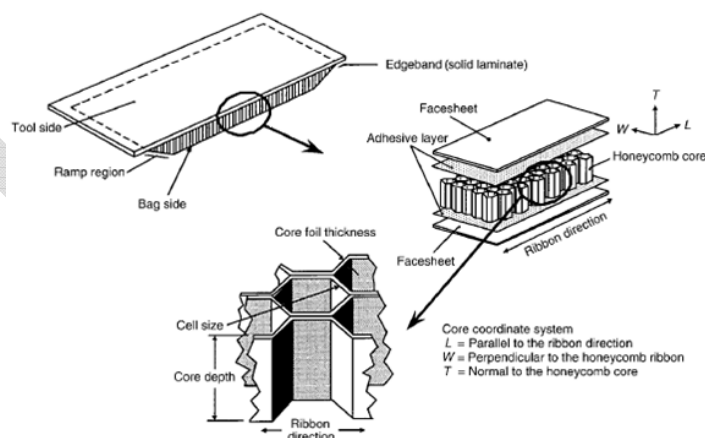


Figure 1. Typical honeycomb sandwich construction (Ref. 3)

A sandwich structure consists of face sheets, core, and some means of connecting the two, such as an adhesive or brazing. The basic core design properties are compressive strength and modulus and shear strength and modulus. The face sheets of a sandwich panel take the bending loads (one face sheet in compression and the other in tension) or in some cases the in-plane shear loads. The major face sheet properties are the compressive, tensile and shear strength and moduli (extracts from CMH-17 Vol. 6).

1.2 SCOPE AND LIMITATIONS

Stress and strength analysis of composite sandwich panels, honeycomb sandwiches in particular.

Internal loads and global buckling calculation methodologies presented in this and the rest of the chapter of this series shall cover only rectangular plane panels with simple load distributions. Other geometries or complex loads states shall be treated by other means (typically numerical FEM analysis).

2 METHOD DESCRIPTION

2.1 INTERNAL LOADS AND STRESS ANALYSIS

2.1.1 INTERNAL LOADS IN THE SANDWICH PANEL

Sandwich beams

The theory for sandwich beams is very similar to the ordinary engineering beam theory but with the addition of shear stresses and transverse shear deformation. This theory is usually referred to as Timoshenko beam theory. The CMH-17, on sub-section §4.5.1, presents some expression to estimate the stiffness and deformations of sandwich beams under certain simple load distributions and support conditions.

Sandwich panels

The stiffness of a sandwich construction can be modelled using classical laminate theory (CLT) for relating in-plane loads N and moments M with the mid-plane strains ϵ and curvature κ by means of the familiar laminates stiffness matrices. Additional terms for computing transverse shear effects can be added.

$$\begin{Bmatrix} N \\ M \end{Bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix} \begin{Bmatrix} \epsilon \\ \kappa \end{Bmatrix} \quad \begin{Bmatrix} Q_y \\ Q_x \end{Bmatrix} = \begin{bmatrix} A_{44} & A_{45} \\ A_{45} & A_{55} \end{bmatrix} \begin{Bmatrix} \gamma_{yz} \\ \gamma_{xz} \end{Bmatrix}$$

Calculation of A, B and D matrices is done similarly to how it is done for fibrous composite laminates, while the in-plane stiffness of the core ply is negligible. On the other hand, for common sandwich arrangements where the transverse shear stiffness of the faces is negligible compared with the transverse shear stiffness of the core...

$$A_{55} = t_c G_{xz}$$

$$A_{44} = t_c G_{yz}$$

$$A_{55} = 0$$

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Transverse and in-plane loadings can be combined using the superposition principle. In case the in-plane loads include compression (and potentially buckling if scaled) the linear superposition of deflections and stresses should be multiplied by the factor:

$$R = 1 / (1 - P / P_{cr})$$

...where P represents the in-plane (compression) applied loads and P_{cr} the buckling load.

As noted above, in sandwich panels the transverse shear stiffness can be of comparable order as bending stiffness and thus, have an impact in internal loads distributions.

(*) However, for initial design phases, simple geometries and loads states, the available solutions for isotropic thin plates could be used, with caution and appropriate transformations, as reasonable approximations for obtaining the critical moments within the panel (required for later stress and local strength analyses). E.g. Estimation of critical bending moment in a rectangular plane sandwich, , with uniform pressure q , from Roarks' Table 11.4 tabulated solution for isotropic plates...

✓ Simply supported along its perimeter (long side a along X direction, short side b)

$$M_y = \sigma_y \cdot t^2 / 6 = (\beta \cdot q \cdot b^2 / t^2) \cdot t^2 / 6 = \beta \cdot q \cdot b^2 / 6$$

...with $\beta = 0.29$ for $a/b=1$, $\beta = 0.61$ for $a/b=2$, ...

✓ If the panel is simply supported with the exception of one free edge the value of β for $a/b=0.5$ is $\beta = 0.36$ (b is the length of the free side, the long one, and the direction of the critical bending plane, not mentioned in table, must be X in this case, parallel to the short side a)

1. Rectangular plate; all edges simply supported

(At center) $\sigma_{\max} = \sigma_b = \frac{\beta q b^2}{t^2}$ and $y_{\max} = \frac{-\alpha q b^4}{Et^3}$

(At center of long sides) $R_{\max} = \gamma q b$

a/b	1.0	1.2	1.4	1.6	1.8	2.0	3.0	4.0	5.0	∞
β	0.2874	0.3762	0.4530	0.5172	0.5688	0.6102	0.7134	0.7410	0.7476	0.7500
α	0.0444	0.0616	0.0770	0.0906	0.1017	0.1110	0.1335	0.1400	0.1417	0.1421

2. Rectangular plate; three edges simply supported, one edge (b) free

$\sigma_{\max} = \frac{\beta q b^2}{t^2}$ and $y_{\max} = \frac{-\alpha q b^4}{Et^3}$

a/b	0.50	0.667	1.0	1.5	2.0	4.0
β	0.36	0.45	0.67	0.77	0.79	0.80
α	0.080	0.106	0.140	0.160	0.165	0.167

Figure 2. Extracts from Roarks (Ref. 4) Formulas for flat plates

Stress analysis of simply supported rectangular sandwich panels and non-uniform external loads is also possible combining energy and numerical (e.g. Rayleigh Ritz) methods. The approach proposed in chapter 3530 "GLOBAL BUCKLING OF COMPOSITE SANDWICH PANELS" , searching for the standard "linear" solutions to the problem instead of the buckling eigenvalues.

For other geometries or more complex loads states the use of dedicated FEM models for obtaining the internal loads is recommended.

2.1.2 STRESS ANALYSIS FROM INTERNAL LOADS

Stresses and strains within the sandwich materials shall be computed from plate/shell loads using the Classical Laminate Theory. See further details in chapter 3510 "STRENGTH OF COMPOSITE SANDWICH PANELS".

2.2 LOCAL STRENGTH

The dedicated methodologies for checking the local strength of honeycomb sandwich, provided the internal loads distribution in the panel is known, are explained in chapter 3510 "STRENGTH OF COMPOSITE SANDWICH PANELS".

2.3 INTERNAL LOADS AND BUCKLING

Methodology for calculating the internal loads and the global buckling onset of a plane rectangular supported sandwich panel can be found in chapter 3530 "RECTANGULAR SANDWICH PANELS".

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
a	mm	Length of a rectangular panel
[A]	N/mm	Stiffness matrix of a composite laminate in in-plane (membrane) loads
b	mm	Width of a rectangular panel
[B]	N	Stiffness matrix of a composite laminate determining the membrane/bending coupling
c	-	As subscript, relative to the core of the sandwich panel, or indication a compression load/stress (from context)
d	mm	Distance between face sheet centroids
[D]	Nmm	Stiffness matrix of a composite laminate in out/plane (membrane & torsion) moments
E	MPa	Elastic Young's modulus
f	-	As subscript, relative to a face of the sandwich panel
G	MPa	Shear modulus
H	mm	Height (thickness) of the core
M	Nmm/mm	Shell out-plane (bending and torsion) moments
N	N/mm	Shell in-plane (membrane) loads
P	N , N/mm	Generic applied load (force, flux, ...)
Q	N/mm	Shell transverse shear load

q	MPa	Pressure on a panel
t	mm	Thickness of a face or thickness of the whole sandwich, from context
X, Y, Z	-	Direction of a local reference axis system, with X and Y parallel to sandwich plane, and X aligned with material reference 0° direction
W	-	As subscript, indicating the transverse direction of a honeycomb core
σ	MPa	Axial stress
τ	MPa	Shear stress

REFERENCES

	Document Reference	Issue	Title
Ref. 1	[SAE] CMH-17	2013	COMPOSITE MATERIALS HANDBOOK. VOLUME 6. STRUCTURAL SANDWICH COMPOSITES.
Ref. 2	[Robert M. Jones]	2 nd Ed.	MECHANICS OF COMPOSITE MATERIALS
Ref. 3	[S. Ward, L. Gintert]	2001	Analysis of Sandwich Structures
Ref. 4	[W. C. Young & R. G. Budynas]	7 th Ed.	ROARKS'S FORMULAS FOR STRESS AND STRAIN
Ref. 5	[C. Kassapoglou]	2 nd Ed.	Design and Analysis of Composite Structures
Ref. 6	NT-T-ADP-0005	Ed. A	MANUAL TEORICO DEL PROGRAMA PANSA

SOFTWARE

Program	Version	Description
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures
PANSA (Airbus DS)	5.5	Stress, strength and buckling analysis of composite sandwich panels
SANDRES (Airbus DS)	0.5	Local strength analysis of composite sandwich panels (prototype/beta non-official version)

Table 1. Software programs referenced in this chapter

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DISTRIBUTION LIST

<i>Department</i>	<i>Name</i>	<i>Full Doc.</i>	<i>Cover Page</i>
Stress Methods & Optimisation (TEASP-TL3)	Dr. Fernaß Daoud	X	
Stress Methods & Optimisation (TEASP-TL3)	Dr. Alexander Hörmann	X	
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
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Structural Repair GET (TEASA-TL6)	Jose Luis Melchor Martin de la Sierra	X	

Table 2. Distribution list (of electronic document, ahead of public release)

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.1 18/06/2018	G. Baños	First issue	All pages

Table 3. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Analysis (TEASA)</i>	Name DD/MM/2018	Name DD/MM/2018
<i>Stress Methods (TEASP-TL3)</i>	G. Baños 18/06/2018	Name DD/MM/2018

Table 4. Approval signatures table for last revision of this chapter